

Interoperable and Open Source Solution for Anchoring Augmented Reality Content to the Real World

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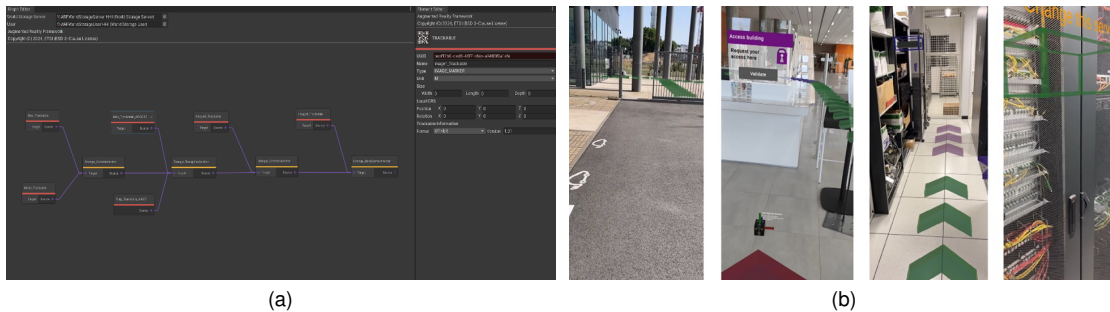


Figure 1: (a) The World Storage editor allows the creation of a graph in Unity containing Trackables and World Anchors of a real environment to which AR assets can be attached. (b) Placements of AR assets relative to the user by the World Analysis component combining data from the World Graph with data provided by an AR runtime or a remote spatial computing platform.

ABSTRACT

This research demonstration presents an open source end-to-end solution for facilitating the anchoring of virtual content in the real world based on various platforms thanks to the interoperable specifications of the ETSI Augmented Reality Framework (ARF) Industry Specification Group (ISG). The solution relies on authoring tools developed for Unity and the created AR applications can be deployed on platforms such as iOS, Android and VisionOS. A validation scenario for data center maintenance is detailed to illustrate the solution. Various examples based on the glTF format are also provided to show the potential of this novel architecture.

Index Terms: Augmented Reality, Interoperability.

1 INTRODUCTION

Augmented Reality (AR) has garnered significant interest due to its potential to enhance user experiences by overlaying digital content onto the real world. However, one of the primary challenges in fully realizing this potential lies in the lack of standardized frameworks for anchoring AR content accurately to physical environments independent of the targeted platform. As an example, OpenXR [2] only supports features such as spatial anchors and image tracking through vendors specific extensions.

The ETSI Augmented Reality Framework (ARF) Industry Specification Group (ISG) proposes a functional architecture [1] that

divides an AR system into several components that communicate through standardized interfaces that have been transposed to OpenAPI interfaces available at <https://forge.etsi.org/rep/arf/openapi>. Examples of components include the World Storage which delivers a representation of the real world, the World Analysis that focuses on the analysis and comprehension of the real world, the AR Authoring tools to prepare the 3D assets of AR scenes, and the Scene Management that manages the scene content according to the user interactions.

In this demonstration, we present an Open Source (Apache v2.0) framework for the Unity game engine based on the ARF specifications that allows digital creators and artists to design virtual content for a given real environment. The process starts with the definition of a World Graph (World Storage component, WS) that provides hierarchical spatial structure of objects of the real world, to which the AR assets will be attached. Then, a World Analysis (WA) component combines at runtime World Graph data with the location data returned by the underlying AR runtime (ARKit, ARCore, etc.) or a remote spatial computing platform to estimate the relative poses between the user and the AR assets. A scenario for data center maintenance illustrates how virtual content can be anchored to the real world based on the use of AR runtime specific features that are seamlessly handled by the WS and WA components. Simple examples based on the glTF format are also provided.

The demonstration at the conference will showcase how to use the tools integrated into Unity, provide examples of simple Trackable augmentations, and present a guidance scenario at the scale of the booth. A video of the demonstration and its details is available here: <https://www.youtube.com/watch?v=wxqSAIa4mVg>.

2 WORLD STORAGE

The World Storage hosts a spatially structured representation of the real world called World Graph, providing all information to register AR assets in 3D. Indeed, to provide 3D registration of AR assets augmenting the user perceptions, the pose of AR devices has to be estimated by matching the signals captured by sensors (e.g., vision, radio waves, or audio) with a priori knowledge of the real environment. This knowledge of the real environment can be represented

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by (i) Trackables (i.e., 2D markers, 3D models, point clouds, radio footprints), elements of the real world whose features are available or can be extracted; and (ii) GeoTrackables, defining position and orientation on earth in a geodetic referential. In addition, World Anchors define fixed positions and orientations in relation to the real world to which AR assets can be attached. Thus, the World Graph links together both Trackables and World Anchors, where each link is a 3D transform defining relative pose between them. Using this graph, the system can estimate the pose between the AR device and a World Anchor by leveraging the Trackables linked to it.

A single World Graph describing a real environment can be used to create different AR experiences (Scenes). To create an AR scene, references to World Anchors have to be added to the Scene Graph, and AR assets can be attached to them. Thus, the World Graph combined with the Scene Graph will be able to provide the 3D transforms between Trackables and AR assets. So, spatial registration of AR assets can be achieved by estimating the pose of the AR device in relation to the Trackables. This is the role of the World Analysis function presented in the next section.

In practice, the World Storage is a remote server accessed through RESTful APIs as defined in the ARF OpenAPI specifications. Source code for storing, manipulating and using World Graphs can be found here <https://labs.etsi.org/rep/arf/world-storage-api-helpers>. Current implementation includes two server implementations, one based on C#/ASP.NET and one based on C++/RestBed. It also includes one package for Unity 2022.3 that proposes a World Graph editor as detailed in Figure 1a. This visual editor allows the construction of the graph structure with Trackables and World Anchors. The pose of these nodes can be visualized in the Unity Scene View window based on a 3D model of the targeted real environment to help their 3D positioning.

3 WORLD ANALYSIS

The World Analysis allows to retrieve the pose(s) of an AR device and/or of one or more World Anchors and Trackables to update the virtual content (AR assets) representation at runtime. These poses are estimated by a localization function (AR runtime such as ARKit or ARCore, or a remote spatial computing platform) based on relocalization information stored in the Trackables provided by the World Storage function. Note that only relocation information compatible with localization function capabilities will be extracted. As an example, if a World Graph contains one World Anchor linked to two Trackables (i.e. an image marker and a mesh), to register AR assets attached to this World Anchor, the World Analysis will compute the pose of the Anchor based on the tracking of the two Trackables (selecting the one which is supported or with the best confidence) and with the relative transform contained in the World Graph between the World Anchor and the Trackables.

In practice, the World Analysis is a remote served accessed through RESTful APIs or a local implementation, both following OpenAPI specifications. Source code for example of World Analysis can be found here: <https://labs.etsi.org/rep/arf/world-analysis-api-helpers>. Current implementation includes an example of local AR Foundation WA implementation compatible with iOS, Android and VisionOS implemented within Unity with AR Foundation. Supported Trackables include (i) image tracker on iOS, Android and VisionOS; (ii) meshes on iOS; (iii) maps on iOS thanks to ARKit ARWorldMap¹ and on Android thanks to ARCore cloud anchors²; and (iv) GeoTrackables on Android thanks to Google GeoSpatial API³. It also includes generation workflows for creating remote World Analysis servers based

¹<https://developer.apple.com/documentation/arkit/arworldmap>

²<https://developers.google.com/ar/develop/cloud-anchors>

³<https://developers.google.com/ar/develop/geospatial>

on C++ or C#/ASP.NET. One example of C# and Python of such server is available in the implementation demonstrating a simulator of remote spatial computing platform. Another example has been implemented in C++ based on the proprietary platform b<>com *Overview*⁴. A Unity package is also available to connect a Unity scene to a local or a remote World Analysis component.

4 EXAMPLE APPLICATIONS

After creating a World Graph with the Unity Editor, the demonstration shows the World Analysis running on an AR device using two example types. The source for these samples is available at <https://labs.etsi.org/rep/arf/world-analysis-api-helpers/unity-world-analysis-validation>.

4.1 glTF AR scenes samples

The repository contains a scene able to load a glTF file that contains 3D content that can be placed according to the World Graph. This aims to facilitate the creation and distribution of new AR content, particularly for designers familiar with this standard format. To do that, a glTF file can include nodes that reference Trackables and World Anchors of the World Graph. Such a node has to respect the following convention "ws:uuid", where *uuid* corresponds to the unique identifier of the corresponding Trackable or World Anchor of the World Graph produced by the WS editor. The end user application then loads the glTF file at runtime while the World Analysis component is responsible to set the pose of these nodes to correctly anchor the 3D content to the real world. As future work, a dedicated extension to the glTF format could be proposed to properly perform this association. Some glTF samples that reference Image Trackables are provided in the repository of these samples.

4.2 Data center guidance and maintenance

To benefit from more advanced rendering and interaction capabilities, Unity scenes can also include game objects that reference nodes of the World Graph. To illustrate this, a validation application for data center maintenance has been developed. The corresponding World Graph and resulting application are depicted in Figure 1a and 1b. This application guides a technician from outside a building to a specific equipment in a server room. It utilizes various types of Trackables like maps, images, or GeoTrackables, as illustrated in the Figure, to ensure compatibilities with multiple platforms. Additionally, it can be adapted to different locations by employing distinct World Graphs, and is compatible with the different World Analysis implementations (local AR Runtime or remote spatial computing platform).

5 CONCLUSION AND FUTURE WORK

This demonstration presents an Open Source pipeline to create AR applications anchored to the real world based on the ISG ARF specifications, using OpenAPI compatible interfaces for communication between the interoperable ARF components. Future work will target the support for additional AR runtimes including WebXR. The authoring part could also be enhanced by allowing the creation and modification of the World Anchors and Trackables directly from the Unity scene view. Immersive tools could even be proposed to perform this authoring.

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⁴<https://b-com.com/en/exploit/overview>